Ghana FX & T-Bill Rate Analysis

Testing the interest-rate to currency transmission mechanism, 2023–2026

Murshida
Portfolio Project

Data: Bank of Ghana, 91-Day Treasury Bill Rate & Interbank GHS/USD Exchange Rate

Weekly observations, 2 January 2023 – 10 August 2026 (189 weeks) · Analysis built in Excel & Power BI

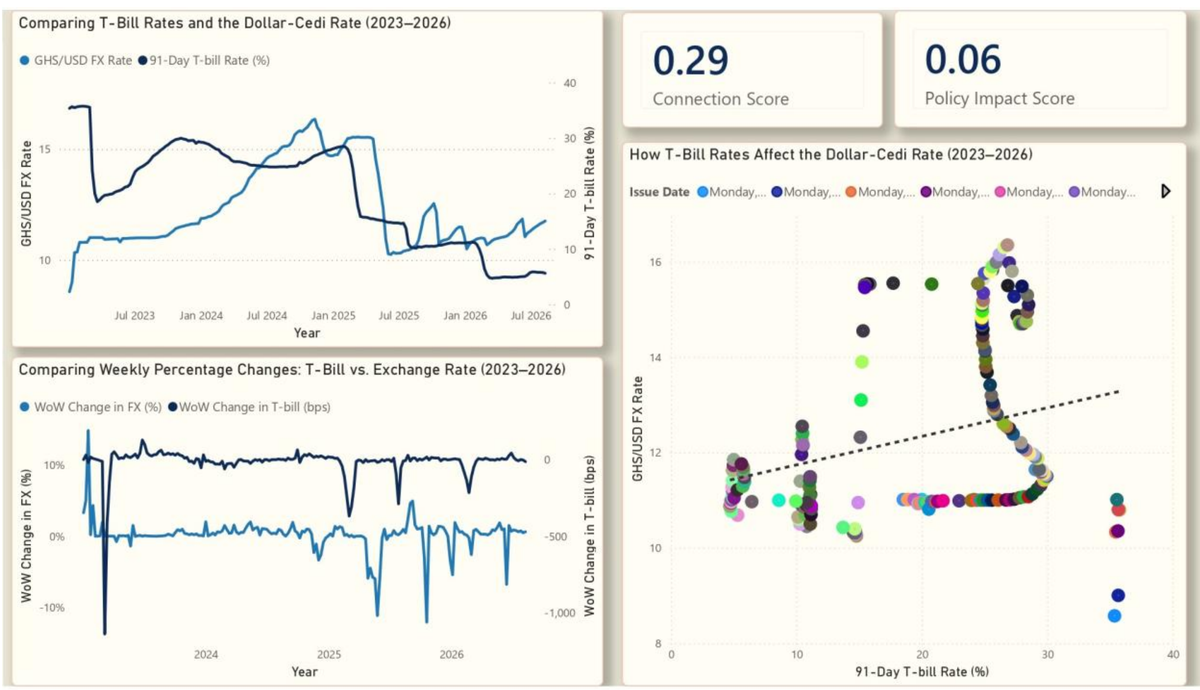

Figure 0. Full Power BI dashboard — rate levels, weekly volatility, and the rate-to-FX relationship.

Contents

1. The Question 2

2. Data & Methodology 2

3. Headline Statistics 2-3

4. Rate and Currency Levels Over Time 3

5. Isolating the Shocks: Week-over-Week Change 3-4

6. The Rate–FX Relationship: Not a Straight Line 5

7. Timeline of Key Policy & Market Events 5-6

8. Key Findings 6

9. Limitations 7

10. Implication: Ghana's Current Easing Cycle 8-9

11. Why This Matters for Global Markets 8

12. Conclusion 8-9

1. The Question

Standard macroeconomic theory holds that when a central bank raises its policy-linked interest rate, it should attract capital into domestic-currency assets and, in turn, support that currency's value. For Ghana, the working hypothesis this project set out to test was straightforward: **does a rise in the 91-day Treasury bill rate actually correspond to a stronger cedi (a lower GHS/USD rate), and does a fall correspond to a weaker one?**

Rather than accepting the theory at face value, this project uses Ghana's own published data across a window that includes a sovereign debt crisis, an IMF programme, and a currency rally — giving the relationship every opportunity to show itself, in either direction.

2. Data & Methodology

Time frame: Weekly data, 2 January 2023 to 10 August 2026 — 189 weekly observations, spanning the 2023 debt restructuring, the 2023 IMF Extended Credit Facility, and the 2025 cedi rally.

Series used: (1) the 91-day Treasury bill interest rate (true yield, not the discount rate, since yield reflects the actual return an investor earns and is the more economically meaningful figure for a capital-attraction hypothesis); (2) the GHS/USD interbank *mid-rate* (the midpoint of buying and selling rates, used because it reflects the market's reference value of the cedi rather than any single bank's transaction spread).

Date alignment: Bank of Ghana's 91-day T-bill auctions are held every Friday, with results officially published the following Monday or Tuesday. To preserve an honest lead-lag reading and avoid look-ahead bias, each week's FX rate was matched to the actual auction Friday — not the later publication date — since that is when the rate was genuinely set by the market.

Derived measures: week-over-week change in FX (percentage, since FX is a price) and week-over-week change in the T-bill rate (basis points, the market convention for a rate that is already a percentage), used to isolate volatility and shocks from the underlying trend.

3. Headline Statistics

0.29	0.06	0.08
Connection Score (Pearson r)	Policy Impact Score (regression slope)	R^2 (variation explained)

Across the full 189-week series, the correlation between the 91-day T-bill rate and the GHS/USD rate is **r = 0.29** — weak, and positive rather than the negative relationship simple theory predicts (higher rates should coincide with a *lower* GHS/USD figure, i.e. a stronger cedi). A simple linear regression of FX rate on T-bill rate produces a **slope of 0.06**: for every 1 percentage-point rise in the T-bill rate, the model predicts the GHS/USD rate moves by only 0.06 — a near-flat relationship in practical terms. The regression's **R^2 of 0.08** means the T-bill rate alone statistically accounts for only around 8% of the week-to-week variation in the cedi's exchange rate; the remaining 92% is driven by factors outside this model.

Regression equation: GHS/USD □ 11.14 + 0.06 × (T-bill rate). At the sample's T-bill mean of 20.0%, this predicts a GHS/USD rate of roughly 12.3 — close to the sample's actual mean of 12.34, confirming the fit is centred correctly even though it explains little of the scatter around that centre.

4. Rate and Currency Levels Over Time

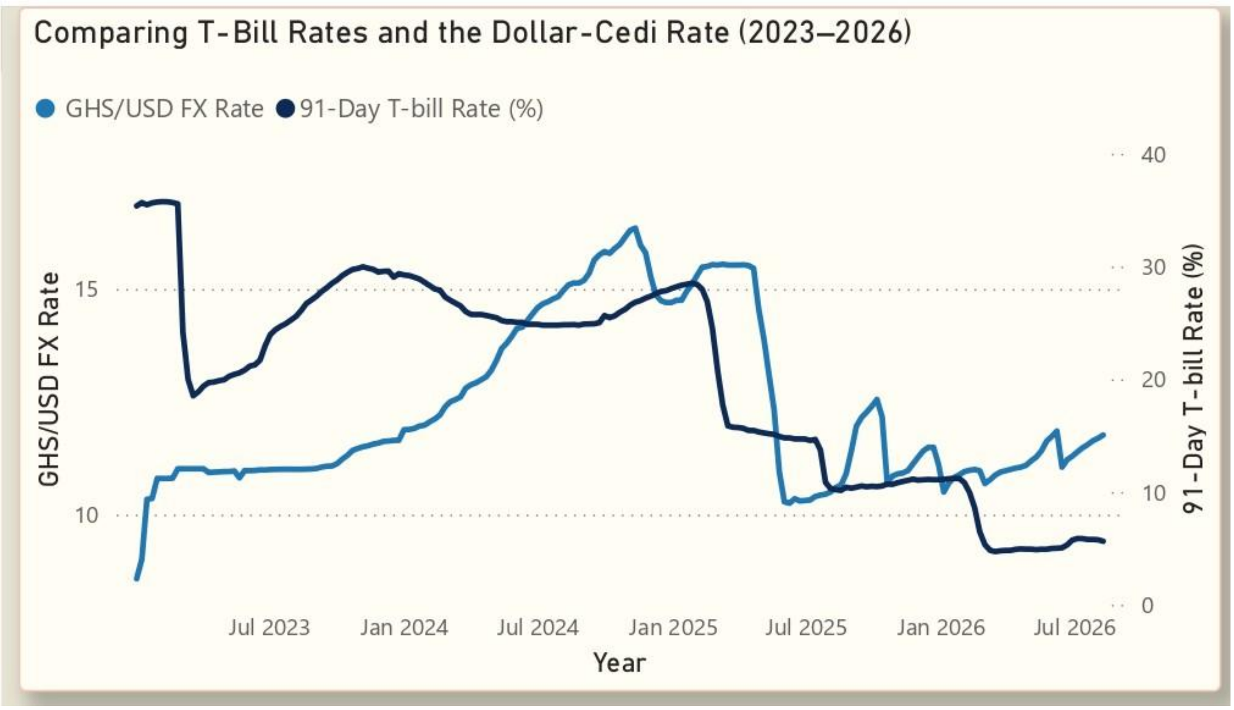

Figure 1. GHS/USD FX rate (left axis) vs. 91-Day T-bill rate (right axis), weekly, Jan 2023–Aug 2026.

The T-bill rate peaked at **35.75%** on 6 February 2023, at the height of Ghana's debt crisis, before falling to a series low of **4.71%** by 16 March 2026 — a decline of over 3,100 basis points across the full window. Over the same period the cedi weakened steadily from an opening rate of **GHS 8.58/USD** (2 January 2023) to a peak weakness of **GHS 16.35/USD** on 11 November 2024, before recovering sharply through 2025 and closing the series at **GHS 11.76/USD** (10 August 2026).

Two distinct regimes stand out. From 2023 through late 2024, the T-bill rate and the cedi's weakness *rose together* — the opposite pairing to what a defensive rate-hike story would predict, consistent with both series being driven by the same underlying debt-crisis pressure rather than one causing the other. From mid-2025 onward, the pattern flips: rates fall sharply while the cedi stabilises and strengthens — again both series moving together, but this time pulled by a common easing/reserve-buildup story rather than a textbook rate-defends-currency mechanism.

5. Isolating the Shocks: Week-over-Week Change

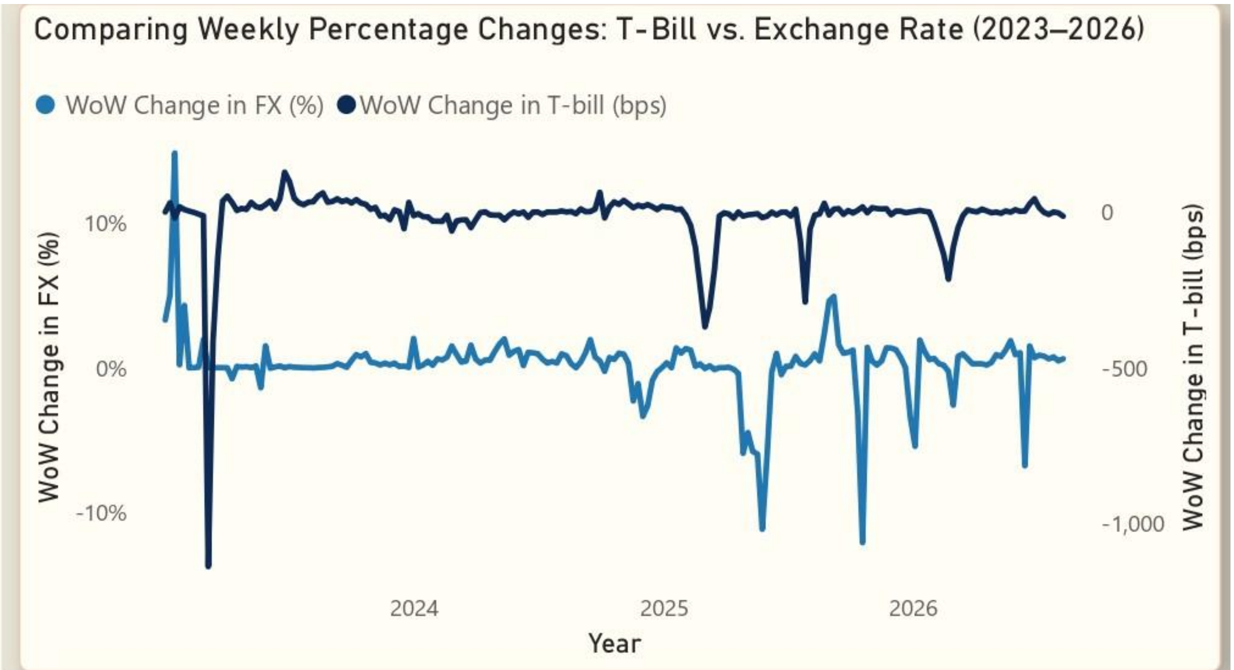

Figure 2. Week-over-week percentage change in FX vs. week-over-week change in T-bill rate (bps).

Looking at week-over-week change rather than levels isolates genuine shocks from gradual drift. One event dominates the entire series: the week of **6 March 2023**, when the T-bill rate fell **1,139 basis points** in a single week — by far the largest move in either series, and roughly three times larger than the next-largest shock (416 bps, the following week). This move coincides precisely with the settlement of Ghana's Domestic Debt Exchange Programme (DDEP) in February–March 2023, which restructured the domestic bond market and mechanically repriced short-term government paper.

A second, smaller shock of **369 bps** appears the week of 3 March 2025, aligning with a surprise Bank of Ghana rate decision made to defend the cedi during that period. Outside these identifiable policy/crisis windows, both series move within a comparatively tight, low-volatility band — reinforcing that the relationship between rates and FX is event-driven rather than a smooth, continuous mechanism.

6. The Rate–FX Relationship: Not a Straight Line

Figure 3. GHS/USD FX rate plotted against the 91-Day T-bill rate, coloured by date, with linear trendline.

Plotting the two series against each other reveals why the correlation is weak: the relationship is not linear, but traces a **loop**. Early observations (2023, T-bill in the low single digits to mid-30s%) sit at the loop's base; as rates climbed through 2023–2024 the cedi weakened in tandem, tracing a path upward and to the right; the 2025–2026 easing cycle then retraces a *different* path back down, at broadly similar FX levels but now paired with much lower rates. A single straight-line regression, and a single correlation coefficient, necessarily average across these two structurally different regimes — which is precisely why both come out weak despite each individual regime showing a much stronger internal pattern.

7. Timeline of Key Policy & Market Events

The T-bill/FX relationship is easiest to read alongside the events that were actually moving the market at each point. The table below lists the major policy and market developments across the sample window, approximately dated to the nearest week in the underlying data.

21 Feb 2023	DDEP domestic bond restructuring settled (~85% bondholder participation)
6 Mar 2023	T-bill rate crashes 35.7% - 24.2% in one week — largest single-week move in the dataset, direct aftermath of DDEP
17 May 2023	IMF Executive Board approves $3bn Extended Credit Facility bailout
29 Jan 2024	BoG's first policy rate cut since 2021 (30% - 29%), as inflation began falling
3 Mar 2025	BoG surprise 100 bps hike to 28%, defending the cedi amid market pressure
28 Apr 2025	Ghana Gold Board (GoldBod) established — mandated gold purchases begin building reserves
28 Jul 2025	BoG cuts policy rate 300 bps (28% □ 25%) as conditions improve
15 Sep 2025	BoG's 126th MPC meeting cuts rate 350 bps to 21.5%
26 Jan 2026	BoG's 128th MPC meeting cuts rate 250 bps to 15.5%
16 Mar 2026	BoG cuts 400 bps to 14% — cumulative 1,400+ bps of cuts since Jul 2025
20 Jul 2026	BoG holds rate at 14% (second hold of 2026)

Note: entries from 2024 onward reflect Bank of Ghana's Monetary Policy Rate (MPR) decisions, which move closely with but are not numerically identical to the 91-day T-bill rate used in the analysis.

8. Key Findings

1. Rates alone do not explain the cedi.

A 1-point rise in the T-bill rate is associated with only a 0.06 move in GHS/USD, and R^2 of 0.08 means T-bill movements account for roughly 8% of the cedi's variation. The remaining 92% is driven by other forces.

2. The relationship is regime-dependent, not stable.

Rates and FX moved together during the 2023 crisis and again during the 2025 rally — but for opposite underlying reasons each time, which a single correlation coefficient cannot capture.

3. One event dominates the whole series.

The Feb–Mar 2023 DDEP settlement produced a 1,139 bps single-week move — by far the largest shock in either series, reflecting debt restructuring, not routine monetary policy.

4. 2025–26 breaks the textbook pattern cleanly.

Bank of Ghana cut the policy-linked rate by over 1,400 bps from mid-2025, while the cedi held broadly stable to stronger — consistent with gold-reserve buildup and Gold Board reforms doing more work than the interest-rate channel.

9. Limitations

• **Weekly, not daily, resolution.** Data is aligned to the weekly T-bill auction cycle. Faster transmission dynamics — a rate move affecting FX within days rather than weeks — would not be visible at this resolution.

• **Correlation and regression are descriptive, not causal.** A weak r and low R^2 show the two series don't move together in a simple, stable way; they cannot rule out reverse causality (FX pressure prompting rate decisions) or the influence of omitted variables such as reserve levels, gold prices, IMF disbursement timing, or broader dollar strength.

• **A single linear model across a regime-dependent relationship.** As Section 6 shows, the rate–FX relationship traces a loop rather than a line — it behaved differently in the 2023 crisis than in the 2025 easing cycle. A single correlation and slope calculated across the full window necessarily blend these two regimes together, understating the strength of the relationship within each one.

• **Data cleaning was manual.** Source data was pulled from the Bank of Ghana's published T-bill and interbank FX releases; one duplicate weekly entry was identified and removed during preparation, and a small number of early-2023 values were spot-checked against the source for plausibility. Minor residual data-entry variance cannot be fully ruled out.

• **Limited crisis cycles observed.** The sample spans one full debt-crisis-to-recovery cycle. A relationship this event-driven is difficult to generalise confidently from a single cycle; a longer or multi-crisis sample would give more confidence in how repeatable these dynamics are.

10. Implication: Ghana's Current Easing Cycle

1,400+ bps	$R^2 = 0.08$	Decoupled
Cut from the policy rate since mid-2025	How little of the cedi's move rates ever explained	Rates fell while the cedi held stable in 2025–26

Bank of Ghana has cut the policy rate by more than 1,400 basis points since mid-2025, and a textbook transmission model would flag that scale of cutting as a currency risk — lower returns should, in theory, push capital out of cedi assets and weaken the cedi. But this analysis shows that relationship barely held even before the cutting cycle began (R^2 of just 0.08), and the 2025–26 period decoupled the two entirely: rates fell sharply while the cedi stayed broadly stable to stronger, carried instead by reserve buildup and gold-export reforms under the new Gold Board.

The current easing cycle likely carries less currency risk than a simple rate model would suggest — provided those offsetting reserve dynamics hold.

11. Why This Matters for Global Markets

Reading a transmission mechanism isn't just plugging numbers into a rule — it's recognising when a relationship is being driven by a common shock rather than a clean cause and effect. That distinction is exactly what separates headline-level macro commentary from the kind of judgement Global Markets work requires, whether the asset in question is the cedi or any other emerging-market currency.

12. Conclusion

This project set out to test a simple, widely-held assumption — that raising interest rates protects a currency — against Ghana's own data rather than accepting it at face value. Across 189 weeks spanning a debt crisis, an IMF programme, and a currency rally, the 91-day T-bill rate and the GHS/USD rate showed only a weak positive correlation ($r = 0.29$), with the rate explaining just 8% of the cedi's movement ($R^2 = 0.08$) — far short of the clean, negative relationship simple theory predicts.

The reason becomes clear once the two regimes are separated: in 2023, rates and FX weakness rose together under debt-crisis pressure; in 2025–26, both fell together as gold reserves and Gold Board reforms took over as the dominant force. Neither period reflects a stable, textbook transmission mechanism — both reflect rates and FX responding to the same external shock, not one driving the other.

The practical takeaway is that Ghana's current, aggressive easing cycle likely carries less currency risk than a simple rate model would suggest, for as long as reserve buildup continues to do the heavier lifting. More broadly, the exercise is a reminder that a single correlation number can mask two very different underlying stories — and that distinguishing between them, rather than defaulting to the textbook assumption, is the more valuable analytical skill.

Source: Bank of Ghana — Treasury Bill Rates & Historical Interbank FX Rates. Analysis and visualisation: Murshida, built in Excel and Power BI.